

23201

MBM UNIVERSITY JODHPUR
DEPARTMENT OF MECHANICAL ENGINEERING

Roll. No. 23VMEC1109

BE IIR IV SEM 2023-24
4ME41A(ME): FLUID MECHANICS

SECOND MID TERM EXAM
TIME 1 HOUR
MAX. MARKS 20

1. A pipe of 15 cm diameter is 14 m long and carries a discharge of 283 litres/s. What loss of head would be saved if central 5 m length of pipe was replaced by 23 cm diameter pipe, the change of section being sudden. Take friction factor as 0.04 for pipes of both diameters. (8)

2. Do the following velocity components represent a possible flow? (4)

$$U = x^3 - y^3, v = z^3 - y^3, w = 3(y^2 - x^2)z$$

3. Write True or False in your copy for the following statements. (8)

- (a) Local acceleration is zero in a steady flow. ✓
(b) Diverging part of a venturimeter is shorter than converging part. ✗
(c) Shear stress in a laminar flow of round pipe remains constant over the cross section. ✗
(d) Pressure gradient in a fully developed pipe flow is constant. ✓
(e) Flow of liquid in conical pipe is always non-uniform. ✓
(f) Pitot static tube measures static pressure only. ✗
(g) In Euler's equation of motion pressure force, gravity force and viscous forces are involved. ✗
(h) Two streamlines can cross each other. ✗

Handwritten calculations:

$$h_1 + h_2 + h_3 = 8.24$$

$$\frac{V_1 - V_2}{2g} = 4.31$$

$$-9.16$$

	9.33	9.33	9.24	9.25	9.32	9.33	9.33	9.34	9.21	9.33
7	9.34	9.32	9.32	9.32	9.33	9.33	9.26	9.24	9.32	9.32
8	9.32	9.34	9.32	9.32	9.24	9.33	9.32	9.32	9.24	9.32
9	9.19	9.17	9.33	9.33	9.26	9.32	9.33	9.33	9.33	9.32
10	9.26	9.35	9.33	9.31	9.34	9.28	9.34	9.33	9.23	9.34

MID TERM-II: - 4ME:42A(ME): METAL CUTTING & METROLOGY

TIME: 1 Hr.

MM: 20

1. What are Autocollimators? Explain. (6)
2. Explain the key measurements and specifications for a screw thread, including the major diameter, minor diameter, effective diameter, best wire size, pitch, and thread angle. (7)
3. In the Analyze phase, project team collected 10 sets of NUT diameter samples (mm) with a subgroup size of 10 from a manufacturing unit. Prepare X-bar and R chart for the given sample of product and comment on the results. For $n=10$, $A_2=0.31$, $D_3=0.22$ and $D_4=1.78$ (7)

S. No.	Measured Values									
	1	2	3	4	5	6	7	8	9	10
1	9.24	9.33	9.33	9.32	9.33	9.25	9.32	9.32	9.33	9.33
2	9.34	9.2	9.33	9.22	9.24	9.23	9.33	9.33	9.33	9.32
3	9.32	9.27	9.19	9.23	9.2	9.33	9.34	9.18	9.32	9.33
4	9.33	9.32	9.32	9.34	9.26	9.33	9.33	9.31	9.24	9.25
5	9.32	9.35	9.34	9.24	9.34	9.37	9.23	9.23	9.33	9.32
6	9.33	9.33	9.24	9.23	9.32	9.33	9.33	9.34	9.27	9.35
7	9.34	9.32	9.32	9.32	9.33	9.33	9.26	9.24	9.32	9.32
8	9.32	9.34	9.32	9.32	9.24	9.33	9.32	9.32	9.24	9.32
9	9.19	9.17	9.33	9.33	9.26	9.32	9.33	9.33	9.33	9.32
10	9.26	9.35	9.33	9.31	9.34	9.28	9.34	9.33	9.23	9.34

DEPARTMENT OF MECHANICAL ENGINEERING
M. B. M. UNIVERSITY, JODHPUR
BE II YEAR IV SEM. MID TERM-2
ME-253A: KINETICS & DYNAMICS OF MACHINES

TIME: 01 HOUR

MM: 30

Q.1 A cam rotating clockwise with a uniform speed of 300 rpm is to give the roller follower of 15 mm diameter the following motion:

- (i) Follower to rise through a distance of 45 mm during 120° cam rotation.
- (ii) Follower to dwell for 60° of cam rotation.
- (iii) Follower to return to its initial position during next 120° of cam rotation and then remain dwell for rest period. The minimum radius of cam is 30 mm and lines of stroke is offset by 15 mm from the axis of the cam and displacement of the follower is to take place with SHM during rise and uniform acceleration / deceleration during return. Draw the cam profile and calculate maximum velocity and acceleration during rise and return.

12 Marks

OR

A Cam with minimum radius of 50 mm, rotating clockwise at a uniform speed, is required to give a knife edge follower the motion as described below:

To move outwards through 40 mm during 100° rotation of the cam.

To dwell for next 80° .

To return to its starting position during next 90° , and

To dwell for the rest period of a revolution i.e. 90° .

Draw the profile of the cam. When the line of stroke of the follower is off-set by 15 mm. The displacement of the follower is to take place with uniform acceleration and uniform retardation. Determine the maximum velocity and acceleration of the follower when the cam shaft rotates at 900 r.p.m.

12 Marks

Q.2 Define the following terms as applied to cam with a neat sketch: (i) Pressure angle (ii) Stroke of the follower (iii) Base circle (iv) Pitch circle

4 Marks

Q.3 In a Porter governor, each of the four arms is 400 mm long. The upper arms are pivoted on the axis of the sleeve whereas the lower arms are attached to the sleeve at a distance of 45 mm from the axis of rotation. Each ball has a mass of 8 kg and the load on the sleeve is 60 kg. What will be the equilibrium speeds for the two extreme radii of 250 mm and 300 mm of rotation of the governor balls?

10 Marks

OR

The mass of each ball of a Proell governor is 7.5 kg and the load on the sleeve is 80 kg. Each of the arms is 500 mm long. The upper arms are pivoted on the axis of rotation whereas the lower arms are pivoted to links of 40 mm from the axis of rotation. The extensions of the lower arms to which the balls are attached are 100 mm long and are parallel to the governor axis at the minimum radius. Determine the equilibrium speeds corresponding to extreme radii of 180 mm and 240 mm.

10 Marks

Q.4 Why is an auxiliary spring used along with main springs in a Wilson-Hartnell governor? Deduce a relationship involving the stiffnesses of these springs and other parameters.

4 Marks

(8)

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BE II Yr. IV SEM.

Machine Design-I

Paper Code- 4ME44A(ME)

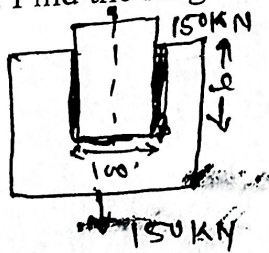
Mid Term Exam.-II

Date: 18/04/24

Max. Marks: 10

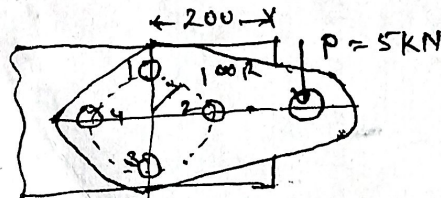
All questions compulsory

Q.1 Two plates joined by means of single transverse and double parallel fillet welds as shown in fig. The size of the fillet weld is 5mm and allowable shear load per mm of weld is 330N. Find the length of each parallel fillet weld.



(3)

Q.2 A riveted joint, consisting of four identical rivets is subjected to an eccentric force of 5 kN as shown in fig. determine the diameter of rivets, if the permissible shear stress is 60 N/mm^2 .



(3)

Q.3. The C- frame of a 100 kN capacity press is shown in fig. The material of the frame is grey cast iron FG 200 and the factor of safety is 3. Determine the dimension of the frame.

(4)

